

CAFFT: Cross-Attention Feature Fusion Transformer for Risky Driving Behavior Recognition

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Abstract—According to the results of a survey administered by the World Health Organization, the primary cause of the majority of traffic accidents is risky driving behavior. Consequently, the identification of risky driving behaviors is imperative for ensuring road safety. In order to effectively address the limitations of existing methods in modeling long-term dependencies, integrating external scenario information, and adapting to different driver personalities, this paper proposes a Transformer-based cross-attention feature fusion model for risky driving behavior recognition, referred to as CAFFT. This model utilizes a hierarchical Transformer encoder structure to capture fine-grained changes and long-term dependencies in driving behavior. Furthermore, the cross-attention mechanism, a cross-modal attention mechanism, is introduced to effectively fuse driving behavior features with weather-related external scenario features, thereby rendering the research scenario more aligned with real-world driving conditions. Finally, through cross-validation and cross-test set generalization experiments, the universality and stability of the method are verified. The experimental results on real driving datasets demonstrate that the CAFFT model achieves substantial performance enhancements in recognizing risky driving behavior. This improvement leads to enhanced accuracy and reliability in recognizing risky driving behavior, thereby providing substantial support for the development of intelligent driving assistance systems.

Index Terms—Risky driving behavior, deep learning, multi-modal feature fusion, situational awareness, driving safety.

I. INTRODUCTION

ROAD safety represents a significant challenge for modern society. The proliferation of vehicles on the road has led to a concomitant increase in traffic accidents, which have become a significant threat to human life and property safety. According to statistics from the World Health Organization (WHO), approximately 1.3 million people die in road traffic accidents each year, with millions more injured [1]. Consequently, how to effectively reduce the incidence of traffic

accidents and improve road traffic safety has become an urgent social issue that requires immediate attention.

Risky driving behavior is a major cause of traffic accidents. Risky driving behavior refers to the behavior of drivers while driving that can lead to traffic accidents, such as speeding, sudden acceleration, sudden deceleration, fatigue driving, distracted driving, drunk driving, etc. [2]. Timely detection and correction of risky driving behaviors by drivers can effectively reduce the likelihood of traffic accidents. However, the identification of risky driving behavior faces the following challenges.

- (a) Driving behavior is a complex temporal process influenced by multiple factors, including driver state, vehicle motion, and the external environment. It is imperative to accurately capture the long-term dependencies and contextual relationships inherent in driving behavior data to effectively identify risky driving behavior [3].
- (b) The identification of risky driving behaviors frequently necessitates a comprehensive evaluation of particular driving circumstances. For instance, even minor speeding may be regarded as risky behavior in inclement weather [4]. How to effectively integrate external situational factors to enhance the accuracy of risky driving behavior recognition remains a challenge.
- (c) Different drivers exhibit divergent driving styles and habits [5]. Consequently, there is a pressing need to develop a universal model for identifying risky driving behavior that can adapt to the individual characteristics of different drivers.

In recent years, the advent of deep learning technology has precipitated novel advances in the identification of risky driving behavior. It has been demonstrated that models such as recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and convolutional neural networks (CNNs) have demonstrated powerful capabilities in processing temporal data and automatic feature extraction [3], [6]. However, these models are still limited in their ability to capture long-term dependencies and integrate multi-source heterogeneous data. The Transformer model has demonstrated remarkable efficacy in natural language processing and computer vision, attributable to its advanced self-attention mechanism [7], [8]. Concurrently, the Transformer model has demonstrated efficacy in capturing long-distance dependencies and exhibiting flexibility in processing multimodal data,

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thereby unveiling novel prospects for recognizing risky driving behaviors.

Based on these, the present paper puts forth a novel Transformer-based model for recognizing risky driving behavior, herein referred to as CAFFT. The model's primary innovations are exhibited in the following domains:

- (a) First, to address the limitations of existing methods in capturing long-term dependencies and fine-grained changes in driving behavior, a hierarchical Transformer encoder structure was designed to obtain more comprehensive local and global feature extraction capabilities for driving behavior.
- (b) Secondly, we introduced a cross-modal attention mechanism, termed Cross-Attention, to effectively integrate driving behavior features with external situational factors (i.e., weather features). This mechanism enhances the perception of driving situations and aligns the research scenarios more closely with real driving scenarios, thereby further improving recognition accuracy.
- (c) Finally, to enhance the adaptability of the recognition method to personalized driver characteristics, cross-validation was employed during training, and the universality and stability of the CAFFT model under various drivers and driving environments were validated through test set generalization tests.

The remainder of this paper is organized as follows. Section II reviews related work, focusing on the latest advances in risky driving behavior recognition, multimodal fusion, and the application of Transformers in time series. Section III describes the task and presents some relevant definitions. Section IV details the architecture and implementation of the CAFFT model. Experimental results on real datasets are shown and discussed in Section V. Finally, Section VI summarizes the paper and discusses future research directions.

II. RELATED WORK

A. Risky Driving Behavior Recognition

There are two main types of data used to identify risky driving behavior. One is image data, which is mainly used to identify distracted and fatigued driving by analyzing images of the driver in the vehicle [9], [10]. The other is time-series data containing driving behavior information, such as trajectory data and driving operation data, which is mainly used to identify risky driving behavior caused by the driver's improper operation of the vehicle, such as sudden acceleration, sudden deceleration, and continuous lane changes [11], [12]. This section focuses on risky driving behavior detection methods based on time-series data. In recent years, researchers have proposed various risky driving behavior detection methods based on time-series data, which can be roughly divided into three categories: statistical classification, traditional machine learning, and deep learning.

Statistical classification methods were among the first methods used to detect risky driving behavior. Such methods typically first extract statistical features from driving behavior data, such as average speed, maximum acceleration, steering wheel rotation frequency, etc., and then determine whether the

driving behavior is risky based on set thresholds. For example, Zhang et al. [13] proposed a statistical rule-based method to identify risky driving behavior by analyzing the behavioral characteristics of drivers, such as speed, acceleration, and braking, to determine whether they are driving while fatigued. This method is simple, easy to understand, and computationally efficient, but it relies on manually designed features and has difficulty capturing complex patterns in driving behavior data.

In order to overcome the limitations of statistical classification methods, researchers began to use traditional machine learning methods for the recognition of risky driving behavior. Machine learning models, including support vector machines (SVM), decision trees, and naive Bayes, have been extensively employed in the domain of risky driving behavior recognition [14], [15], [16]. For instance, Yu et al. [14] proposed an SVM-based model for identifying risky driving behavior. This model analyzes behavioral characteristics such as the driver's speed, acceleration, and steering wheel rotation to determine whether the driver is in a dangerous driving state. Compared with statistical classification methods, traditional machine learning methods can automatically learn patterns in driving behavior data and achieve higher recognition accuracy. However, traditional machine learning models still rely on manually designed features and have difficulty processing high-dimensional, nonlinear driving behavior data.

The advent of deep learning technology has prompted researchers to adopt deep learning models for the recognition of risky driving behavior. Deep learning models, including recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and convolutional neural networks (CNNs), have been extensively employed in the domain of risky driving behavior recognition [3], [6]. For instance, Yang et al. [17] proposed a behavior prediction model, comprising a CNN and a two-layer LSTM, which integrates vehicle lane-change feasibility and lateral offset rate of change. This model effectively identifies high-risk lane-change behaviors. A comparison of deep learning-based models with traditional machine learning methods reveals that the former can automatically learn high-dimensional, nonlinear features in driving behavior data, achieving higher recognition accuracy. Nevertheless, these models continue to exhibit limitations in their capacity to manage long-range dependencies.

B. Multimodal Fusion

In real-world scenarios, information from a single source frequently exhibits limitations, failing to satisfy the system's requirements for perception capabilities and decision-making precision. Consequently, the integration of data from diverse modalities and sources to construct more comprehensive and stable systems has emerged as a pivotal area of research interest. Multimodal data fusion is the key technology to address this issue, aiming to integrate data from different modalities and leverage their complementarity and redundancy to enhance the overall system performance [18]. Multimodal data fusion has been widely applied in various fields, such as image captioning [19] (combining image content with textual descriptions), video understanding (fusing visual, auditory,

and textual information), and sentiment analysis (combining speech intonation with textual content) [20], [21], [22].

Similarly, risky driving behavior is shaped not only by driver state but also by scenario-level contextual factors and uncertainty. Therefore, the effective integration of multi-source heterogeneous data, such as vehicle driving data, environmental perception data, and driver physiological data, is crucial for improving recognition accuracy.

In recent years, multi-modal data fusion technology has seen widespread application in the recognition of risky driving behavior. For instance, certain researchers have integrated vehicle speed, acceleration, and other driving data with environmental data, such as weather conditions and road conditions, to enhance the model's adaptability to complex driving scenarios [23], [24]. Concurrently, other researchers have endeavored to integrate driver physiological data, including facial expressions and eye movements, with vehicle operation data to enhance the precision of identifying driver fatigue or distracted driving [25]. For instance, Wang et al. [26] investigated three distinct human-machine interaction (HMI) modes—control, warning, and guidance—and discovered that varying information presentation significantly alters drivers' acceleration behavior, strategy selection, and merging position in highway merging scenarios. This directly substantiates the substantial impact of the external interactive environment as a contextual factor on driving behavior.

According to different fusion strategies, existing multimodal risky driving behavior recognition methods can be roughly divided into the following categories: Based on early fusion methods, Haris and Glowacz [27] spliced vehicle driving data and environmental information, and then input them into a deep learning model for training. Based on late fusion methods, Liu et al. [28] extracted vehicle trajectory features and environmental factors such as weather, and then fused them by weighted averaging. Based on the attention mechanism, Mou et al. [29] proposed a multi-modal fusion method based on the attention mechanism to detect driver fatigue.

While multi-modal data fusion methods have demonstrated progress in identifying risky driving behavior, the relative importance of each data source varies across different driving scenarios. Existing methods do not incorporate situational factors as important factors affecting driving behavior into the model, making it difficult to maintain good performance in complex and changing real driving environments. Concurrently, the prevailing methodologies predominantly emphasize the integration of static features, while disregarding the recognition that driving behavior is inherently sequential. This oversight results in an inability to capture the temporal dependencies inherent in driving behavior data, consequently hindering the discernment of dynamic variations in driving behavior.

C. Transformers in Time Series

The Transformer model was first proposed by Vaswani et al. [30]. This model has been instrumental in the field of natural language processing (NLP) due to its pioneering introduction of self-attention mechanisms and an encoder-decoder architecture. In contrast to conventional

recurrent neural networks (RNNs), the Transformer model relies exclusively on self-attention mechanisms to process sequence data in parallel, thereby effectively capturing long-range dependencies. The self-attention mechanism enables the system to discern associations between disparate positions in a sequence, thereby facilitating a comprehensive perception of sequence information. Due to the high computational complexity of the self-attention mechanism, researchers have proposed various improved self-attention mechanisms, such as sparse attention [31] and window attention [32].

In recent years, transformer models have demonstrated notable advancements in time series tasks, including time series prediction, anomaly detection, and classification [33], [34]. A comparison of Transformer models with traditional time series models (e.g., ARIMA and LSTM) reveals their superiority in capturing long-term dependencies and enhancing computational efficiency through parallel computing. In the field of time series prediction, Li et al. [35] have proposed a Transformer-based time series prediction model that has been shown to improve prediction accuracy by learning the long-term trends of time series data. Zhou et al. [31] proposed a Transformer-based time series prediction model that reduces computational complexity and improves the efficiency of long-sequence prediction by introducing the ProbSparse self-attention mechanism. In the context of time series anomaly detection, Tuli et al. [36] have proposed a Transformer-based anomaly detection model that identifies abnormal events by learning the patterns of normal time series.

Despite the noteworthy advancements of Transformer models in time series tasks, their utilization in the domain of risky driving behavior recognition remains a relatively recent development. In a recent study, You et al. [37] used Transformer models to learn feature representations in driving behavior data and input driving behavior sequences into Transformer models to achieve driving behavior classification. Wang et al. [38] proposed a novel approach that integrates the transformers and convolutional neural networks (CNNs) to generate input vector representations of driving behavior. These representations are then fed into a classifier, with the objective of identifying whether the driver is distracted. Zhang et al. [39] developed an effective method for extracting features from input images using Swin Transformers. They then enhanced the discriminative power of different categories in the feature space, achieving high accuracy in distracted driving behavior recognition tasks.

However, the majority of extant Transformer-based methods for recognizing risky driving behavior, such as Informer and the Temporal Fusion Transformer (TFT), tend to treat driving behavior data as a whole, thereby overlooking the fine-grained temporal fluctuations inherent in driving behavior. Concurrently, divergent assessments of driving peril may emerge under different driving scenarios and external conditions. These general-purpose Transformer architectures primarily focus on long-range dependency modeling or temporal feature selection within a single modality, without explicitly modeling scenario-dependent modulation of driving behavior. As a result, existing methods struggle to adequately account for situational variability across different scenarios and are

difficult to adapt to the heterogeneous requirements of real-world risky driving behavior recognition.

III. SYMBOLS, DEFINITIONS, AND TASK DESCRIPTION

The dataset we used encompasses three distinct sets of time-related data related to vehicles, namely speed, acceleration, and angular velocity. In addition, the dataset incorporates situational factors relevant to driving behavior, such as weather information. In this section, we delineate several of the definitions and symbols employed in the risky driving behavior recognition model. Furthermore, we describe the task of recognizing risky driving behavior by integrating situational information.

A. Symbols and Definitions

Definition 1 (Trip P): A trip P represents the driving trajectory data of a vehicle from a dataset during the entire driving process. It consists of the state P_t of the vehicle at multiple instantaneous points. The state P_t of the vehicle at each instant includes speed information v_t , acceleration information a_t , angular velocity information ω_t , and longitude and latitude information. $V \in R^T$, $A \in R^T$, and $W \in R^T$ are the continuous speed information, acceleration information, and angular velocity information during the entire trip P , respectively. T represents the time spent by the vehicle during trip P , representing the length of the time series. The interval between every two timestamps is one second.

Definition 2 (Driving Process D_P): The driving process D_P is represented by the driving status of the vehicle within the selected sliding time window (the size of the sliding time window is L) for L seconds, i.e., $D_P \subseteq P$. Then, trip P encompasses $N = (T - L)/S + 1$ (S is the sliding step length) driving processes.

Definition 3 (Situational Factors SF): Situational factors (SF) refer to situational environmental factors that affect vehicle driving behavior, mainly considering weather information. It includes information such as weather type, wind direction, wind speed, and temperature. Based on the different data types, the One-Hot encoding method and normalization method are used to process this information to obtain a weather data matrix $C \in R^{T \times D}$ that can be processed by the model (T is the length of the time series and D is the weather data dimension at each moment).

B. Task Description

The task of identifying risky driving behavior is described below. Input: Time series containing vehicle trajectory information (time information t , speed information v_t , acceleration information a_t , angular velocity information ω_t , and longitude and latitude information); situational factors SF (weather data matrix C) that affect driving behavior recognition. Output: Risk driving behavior recognition results $\hat{P}$, which include all risky driving behaviors during the entire trip P .

IV. METHODS

As illustrated in Fig. 1, the CAFFT model is comprised of three primary modules: a driving sequence feature learning

module, a weather feature learning module, and a cross-modal dynamic fusion module.

Due to its excellent performance in capturing long-range dependencies and interactions, Transformer is well suited for time series analysis. Consequently, the driving sequence feature learning module is comprised of a hierarchically designed Transformer encoder. This module not only fully captures the fine-grained information of driving behavior but also considers the long-term dependency characteristics of driving behavior. Additionally, it has been designed to address the issue of misjudgment and omission in assessing driving behavior risk. Specifically, we use Transformers to extract historical driving state information as basic features. Incorporating the self-attention mechanism, we integrate driving process features under varying window sizes, thereby complementing and enhancing each other. This approach prevents misjudgment and the omission of driving behavior risks caused by the division of the driving process, thus enhancing the accuracy of feature extraction. In the weather feature learning module, an effective extraction method was designed to identify weather features related to driving behavior. This method combines One-Hot encoding, numerical normalization, and a one-dimensional convolutional network. Furthermore, in the cross-modal dynamic fusion module, we integrate driving behavior features with weather features through a cross-attention mechanism to characterize the impact of situational information on driving behavior during the driving process. This approach aligns more closely with real driving scenarios and facilitates the identification of risky driving behavior in the real world.

A. Data Preprocessing and Window Alignment

To effectively identify risky driving behaviors within trajectory data, we first performed outlier detection and removal from the raw dataset. Specifically, we excluded instances where vehicles were stationary for extended periods, anomalous readings caused by GPS signal drift, and brief movements insufficient for driving behavior analysis. Subsequently, a team comprising data analysts and data quality coordinators manually verified and analyzed the trajectory data files for annotation.

“High-risk” driving behavior was defined as the presence of one or more of the following four conditions: overspeeding, abrupt longitudinal maneuvers, sharp turns, and fatigued driving. Specifically:

- (a) **Overspeeding:** Vehicle speed exceeding 80 km/h or 40 km/h (in foggy conditions) for a duration greater than 3 seconds.
- (b) **Abrupt Longitudinal Maneuver:** An instantaneous acceleration $\geq 0.3\mu g$ or $\leq -0.3\mu g$ occurring between adjacent sampling points within a 3-second interval, where the friction coefficient μ is inferred from aligned weather data, with $\mu \in \{0.9(\text{clear weather}), 0.5(\text{rain/snowy weather})\}$.
- (c) **Sharp Turn:** Angular velocity $> \pi/6$ rad/s, or turning approaching the grip limit—defined as $v > \sqrt{\mu g R}$ where $R = v/\omega$. The value of μ is consistent with that defined previously.

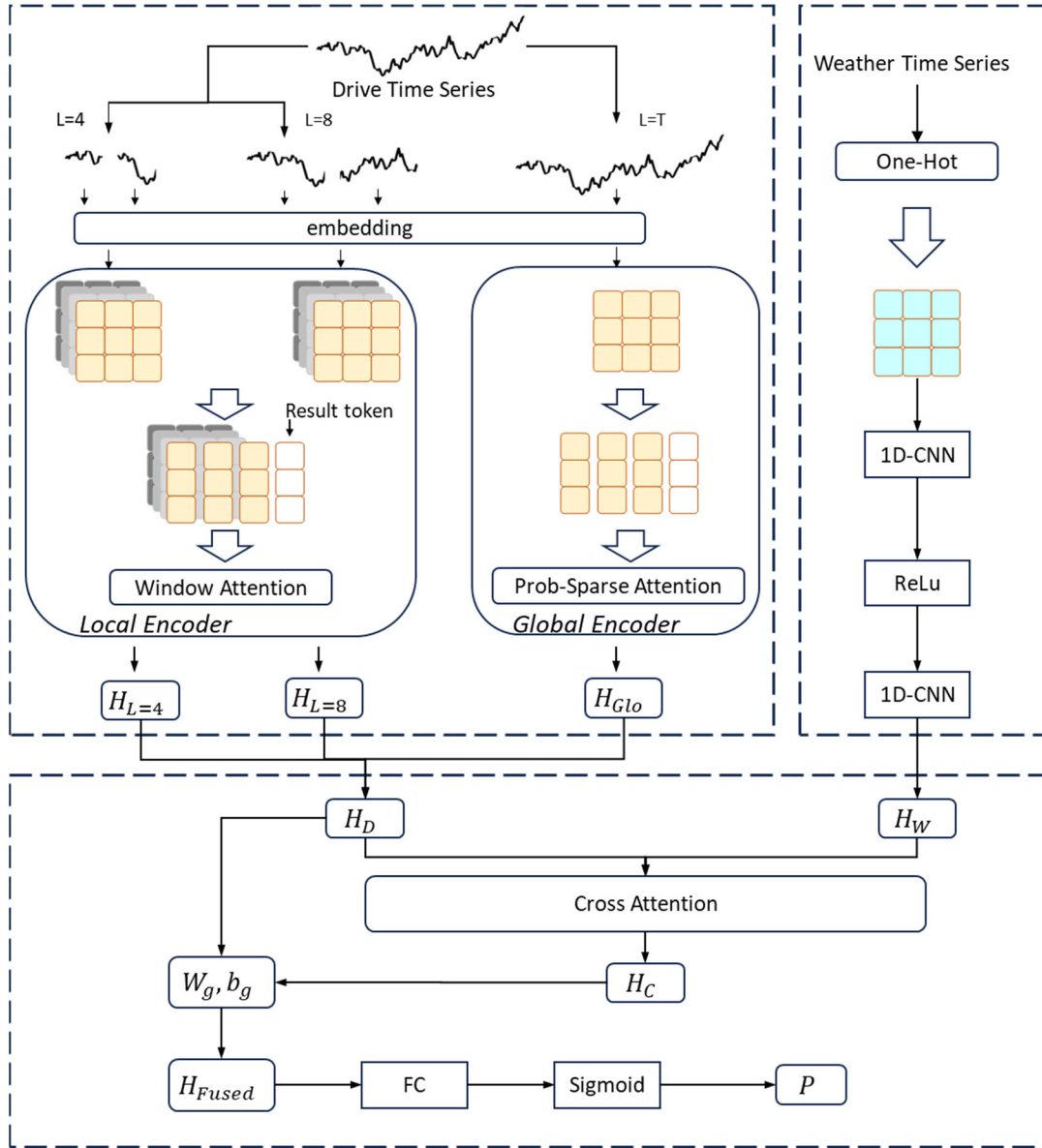


Fig. 1. Overall structure of our proposed CAFFT model. The upper left panel illustrates the Driving Temporal Feature Learning module, which hierarchically extracts trajectory features. The upper right panel depicts the Weather Feature Learning module. The lower panel presents the Cross-modal Dynamic Feature Fusion module, responsible for integrating trajectory and weather features.

- (d) **Fatigued Driving:** Continuous driving duration exceeding 4 hours without an uninterrupted rest period greater than 20 minutes. All driving data beyond the 4-hour mark are subsequently flagged as fatigued driving.

Given the differing sampling rates between trajectory and weather data—hourly for weather data and 1 Hz for trajectory data—we adopted a midpoint alignment strategy to synchronize the time windows of these datasets. This aligns the hourly weather data with the second-level trajectory timeline. Let the hourly weather data interval be denoted as $H_i = [h_i, h_{i+1})$, and the trajectory time window as $W = [t_s, t_{s+L}]$, where L is the window length. The division of time windows is detailed in Section IV-B, pertaining to driving temporal feature learning. The window midpoint is defined as:

$$t_m = \frac{t_s + t_{s+L}}{2} \quad (1)$$

The weather data vector for this window is then assigned the measured value corresponding to the hour to which the midpoint belongs:

$$\mathbb{W}_W = \mathbb{W}_{hour(t_m)} \quad (2)$$

For detailed descriptions of different weather data types and their values, please refer to the section on weather feature learning. Since the selected window length is significantly less than one hour, the vast majority of windows fall entirely within a single hourly period. In the extremely rare event that a window spans an hourly boundary, our convention assigns it to the subsequent hourly period using a half-open interval definition. This strategy avoids inappropriate interpolation smoothing of discrete variables and maintains alignment consistency and reproducibility across different window settings.

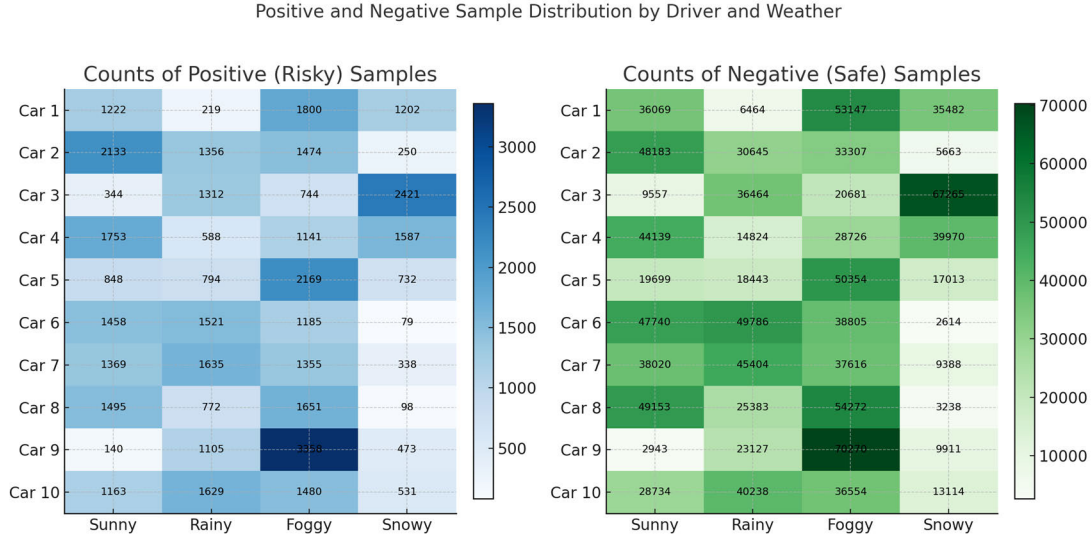


Fig. 2. Distribution of positive and negative samples per “driver-weather”. The left figure shows the distribution of risky driving behavior samples, and the right figure shows the distribution of safe driving behavior samples.

Following the aforementioned data preprocessing, we obtained 6851 trajectory data points, with an average of approximately 15 trip records per vehicle. Each trajectory data point comprised approximately 8354 entries, with each trip lasting about 2.3 hours. A total of 2,113,854 data points were labeled as “high-risk,” accounting for approximately 3.69% of the total dataset. Furthermore, we selected 10 vehicles from 450, representing various regions and operating timeframes, to illustrate the distribution of positive and negative samples of risky driving behavior under different driver and weather conditions. The statistical results are presented in Fig. 2. In the figure, each cell represents the number of samples for a specific “driver-weather” combination, with darker colors indicating a higher sample count.

Although the processed dataset exhibits a severe class imbalance, with only approximately 3.69% of samples labeled as high-risk, no explicit class re-balancing techniques such as class-weighted loss, focal loss, or over-/under-sampling were employed. Instead, this study addresses the imbalance primarily at the evaluation and decision-making level by adopting recall-oriented metrics ($F\beta$ with $\beta > 1$), PR-AUC, and threshold sensitivity analysis. This design choice is motivated by the safety-critical nature of risky driving behavior recognition, where suppressing false negatives is more important than achieving balanced class distributions during training. The effectiveness of this strategy is further supported by statistical significance testing and consistent performance across cross-test set generalization experiments.

B. Driving Sequence Feature Learning

Indeed, risky driving behavior is a continuous process that can vary in duration. Consequently, a single-grain driving process extraction is inadequate for accurately describing all driving behaviors. As demonstrated in Fig. 3, when a limited time window is selected, the resulting driving behavior may be deemed safe. However, when an extended time window is

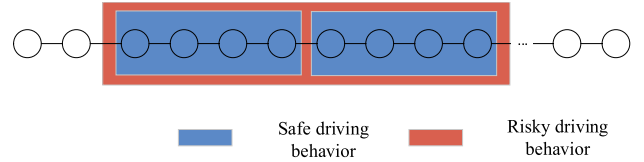


Fig. 3. Misjudgment and omission of driving behavior risks. The box represents driving behavior within the time series.

considered, the driving behavior may be identified as risky and vice versa. Therefore, this paper proposes a hierarchical Transformer encoder structure to capture the fine-grained fluctuations and long-term dependencies of driving behavior. In the local encoder, multi-granularity driving behavior time series segmentation is employed to extract and learn the driving process from the preprocessed driving trajectory data. The global sparse attention of the global encoder is utilized to capture the global dependencies of the driving process and to mine the correlation between multiple driving processes.

To effectively extract features from vehicle trajectory data, as illustrated in Fig. 4, the trajectory data (i.e., $V \in R^L$, $A \in R^L$, and $W \in R^L$) is segmented and z-score normalized to obtain a matrix $Y \in R^{L \times 3}$. On this basis, Y is further projected from three dimensions to d dimensions of feature space through a fully connected layer to obtain a feature sequence $X \in R^{N \times L \times d}$, providing richer feature information for subsequent driving behavior feature learning.

In the feature extraction of the driving process, in order to extract the temporal features of the driving process, a hierarchical Transformer structure was designed to capture the fine-grained fluctuations and long-term dependencies of driving behavior. This structure contains two main encoder layers, the Global Encoder and the Local Encoder.

1) *Global Encoder*: In order to capture long-term dependencies in driving behavior, a global encoder is introduced. The global encoder is a global consideration of the entire trip,

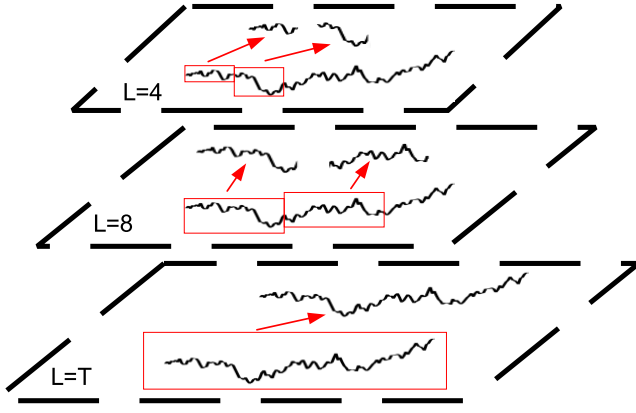


Fig. 4. Multi-granularity division of driving behavior. Different window sizes represent varying granularities of division.

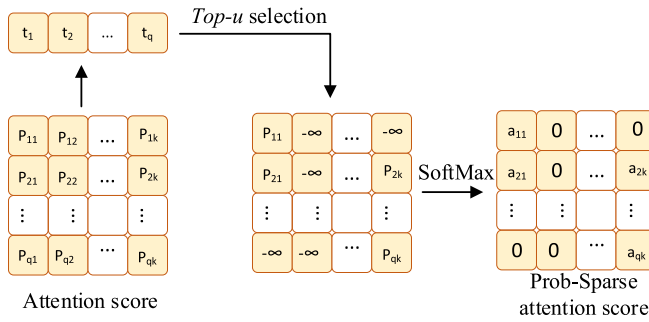


Fig. 5. The illustration of the Prob-sparse attention. Exclusively retains the *Top-u* queries with high variance.

so the window size $L = T$ and the number of windows $N = 1$. At this point, the feature sequence is $X \in R^{T \times d}$. At the same time, in order to better store the results of feature learning, a blank result token is added to the feature sequence X (refer to the processing method in [40]). Consequently, the input of the global encoder is the sequence $X_{Glo} \in R^{(T+1) \times d}$.

The global encoder employs Prob-Sparse Attention [31], an approach designed to minimize computational complexity and accentuate salient information. The calculation formula for Prob-Sparse self-attention is as follows:

$$\begin{aligned} & \text{ProbSparseAttention}(Q_G, K_G, V_G) \\ &= \text{Softmax} \left(\frac{\bar{Q}_G K_G^T}{\sqrt{d_{k_G}}} \right) V_G \end{aligned} \quad (3)$$

where $Q_G, K_G, V_G \in R^{(T+1) \times d}$ represent the Query, Key, and Value matrices, respectively, and d denotes the embedding dimension. $\bar{Q}_G$ is the sparse query matrix, which exclusively retains the *Top-u* queries with high variance (Fig. 5). This sparsification strategy effectively reduces the number of attention weights that need to be calculated and improves computational efficiency. In particular, the variance of each query vector is calculated, and the *Top-u* query vectors with the highest variance are selected. These vectors are considered to best represent the key information of the input sequence. The output of the global encoder is the global feature $H_{Glo} \in R^{1 \times d}$, which reflects the long-term state of driving behavior.

2) *Local Encoder*: Additionally, to capture the fine-grained fluctuations in driving behavior, a local encoder has been introduced. To learn driving behavior features more comprehensively, the input to the local encoder is divided into different granularities. That is to say, the time series is divided using sliding time windows of different sizes. Similar to the global encoder, to better store the results of feature learning, a blank result token is added to each window. Therefore, the input to the local encoder is sequence $X_{Loc} \in R^{N \times (L+1) \times d}$, where N signifies the number of windows divided into different granularities, and L denotes the size of windows of different granularities.

The Window Attention mechanism [32] is adopted for the extraction of window features, as it calculates attention weights exclusively within the window. The calculation formula for Window self-attention is as follows:

$$\text{Attention}(Q_L, K_L, V_L) = \text{Softmax} \left(\frac{Q_L K_L^T}{\sqrt{d_{k_L}}} \right) V_L \quad (4)$$

where $Q_L, K_L, V_L \in R^{(L+1) \times d}$ represent the Query, Key, and Value matrices, respectively, and d denotes the embedding dimension. Local window attention facilitates the model's efficient learning of correlations between time steps within a window, thereby reducing computational complexity. Within each window, time steps share position encoding to distinguish between different time steps within the window. In the context of the granularity division of sliding time windows ($L = 4, S = 4$), the local encoder's output is defined as the local feature $H_{L=4} \in R^{N_1 \times 1 \times d}$. Similarly, under the granularity division of sliding time windows ($L = 8, S = 8$), the local encoder's output is defined as the local feature $H_{L=8} \in R^{N_2 \times 1 \times d}$. It is noteworthy that the number of windows under the corresponding granularity division is denoted by N_1 and N_2 . It captures the short-term dynamics of driving behavior at different granularities.

3) *Feature Fusion*: In order to ensure the effective utilization of local and global features of varying granularities, local features $H_{L=4} \in R^{N_1 \times 1 \times d}$, $H_{L=8} \in R^{N_2 \times 1 \times d}$, and global features $H_{Glo} \in R^{1 \times d}$ are replicated to align the feature dimensions:

$$H'_{L=8} = \begin{bmatrix} H_{L=8} \end{bmatrix} \quad (5)$$

$$H'_{Glo} = \text{tile} [H_{Glo}, [N, 1]] \quad (6)$$

This process yields $H'_{L=8} \in R^{N \times d}$ and $H'_{Glo} \in R^{N \times d}$. Additionally, driving behavior characteristics H_d are derived by establishing a connection between the residuals. The following calculations are to be performed:

$$\begin{aligned} H_D &= LN(H_{L=4} + H'_{L=8} + H'_{Glo} \\ &\quad + FFN(H_{L=4} + H'_{L=8} + H'_{Glo})) \end{aligned} \quad (7)$$

where $LN(\text{LayerNorm})$ denotes the process of normalization, while FFN represents the Feed-Forward Network. The FFN increases the feature dimension from d to $d1$ through the initial linear layer, and subsequently reduces the dimension back to d through the subsequent linear layer. This fusion method retains the detailed information of driving behavior while also considering its long-term dependencies, providing a more comprehensive feature representation for subsequent risky driving behavior recognition.

C. Weather Feature Learning

In order to effectively extract weather features H related to driving behavior, a feature extraction method was designed that combines One-Hot encoding, numerical normalization, and one-dimensional convolutional networks(1D-CNN). This section will provide a detailed exposition of the preprocessing of meteorological data and the extraction of features.

Due to the different characteristic attributes of weather types, wind direction, wind speed, and temperature, different encoding strategies were adopted to better adapt to the subsequent one-dimensional convolutional network processing.

1) **Feature Encoding: Weather type:** Weather type is a discrete variable. To enable it to be processed by the model, it is encoded using the One-Hot encoding method. We simplify weather types into four distinct categories (e.g., sunny, hazy, rainy, snowy). Each weather type is encoded as a sparse binary vector of length 4, where the element corresponding to the weather type is 1, and all other elements are 0. This encoding effectively represents the category information of weather types.

Wind direction: As with the classification of weather types, the direction of wind is encoded through the implementation of the One-Hot encoding method. The eight wind directions are represented by the following letters: east, west, etc. up to northeast and northwest. The direction of the wind is encoded as an 8-length sparse binary vector.

Wind strength: Wind strength is also a discrete variable, and it is encoded using the One-Hot encoding method. According to the national standard “Wind Force Scale” published by China in 2012, wind strength is divided into 18 levels [41]. However, given that the experimental data utilized in this study excludes wind speeds above level 10, the wind strength is encoded as a sparse binary vector of length 11.

Temperature: Temperature is a continuous numerical variable. To promote the acceleration of model convergence, the normalization of the temperature data is employed. Specifically, the temperature data is linearly scaled to the interval [0, 1]. The normalization formula is as follows:

$$\mathbb{T}_{norm} = \frac{\mathbb{T} - \mathbb{T}_{min}}{\mathbb{T}_{max} - \mathbb{T}_{min}} \quad (8)$$

where $\mathbb{T}$ denotes the original temperature value, $\mathbb{T}_{min}$ and $\mathbb{T}_{max}$ denote the minimum and maximum temperature values, respectively, and $\mathbb{T}_{norm}$ denotes the normalized temperature value. This normalization method effectively maps temperature data to a unified numerical range, thereby improving the training efficiency and stability of the model.

After the above encoding and normalization, the hourly weather data can be represented as a feature vector $C_t \in R^D$ with dimension $D = 4 + 8 + 11 + 1$, where the first 4 + 8 + 11 elements represent the One-Hot encoding of weather types (4), wind direction (8), and wind speed (11), and the last element represents the normalized temperature value. Combine all weather data at all times to obtain the weather data matrix $C \in R^{T \times D}$, where T is the total time of the corresponding trip.

2) **1D-CNN:** Next, weather features are extracted from the preprocessed weather data matrix $C \in R^{T \times D}$ using 1D-CNN.

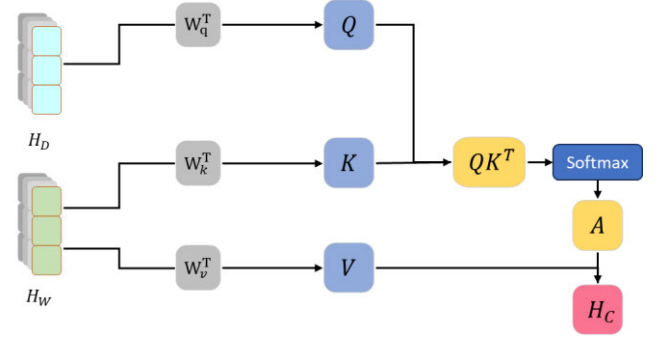


Fig. 6. Cross Attention work mechanism. Cross-modal fusion is performed using cross-attention, with driving features H_D as the query and weather features H_W as the key-value pairs.

Convolutional Layer: Two one-dimensional convolutional layers are employed to capture local correlations in weather sequences. It has been established that each convolutional layer utilizes the same configuration, which consists of a convolution kernel size of 1 and a stride of 1. The formula for the convolution operation is as follows:

$$H_{conv} = \text{Conv1D}(\text{ReLU}(\text{Conv1D}(H_{conv}))) \quad (9)$$

The symbol Conv1D denotes a one-dimensional convolution operation, and the symbol ReLU denotes the ReLU activation function. The ReLU activation function has been demonstrated to effectively alleviate the vanishing gradient problem and accelerate model training. The number of output channels for the two convolution layers is $d/2$ and d , respectively. A gradual increase in the number of channels can facilitate the extraction of richer feature information.

The final step in the process entails the interpolation of the feature vector $H_{conv} \in R^{T \times d}$, which is obtained through the convolution operation, to obtain the weather feature $H_W \in R^{N \times d}$, Where N numerically equal to the number of windows obtained during the driving behavior feature learning process.

Through the above-mentioned mixed feature encoding and one-dimensional convolution network, weather features related to driving behavior can be effectively extracted, providing reliable input for subsequent cross-modal fusion and risky driving behavior recognition.

D. Cross-Modal Dynamic Fusion

In order to correlate the interaction between driving behavior and weather, a dynamic fusion mechanism was designed to achieve effective fusion of driving behavior features and weather features.

Initially, a cross-attention mechanism (Fig. 6) was employed to calculate the correlation between driving features and weather features. The driving feature H_D was used as the Query, and the weather feature H_W was used as the Key-Value.

Cross-modal correlation was then calculated:

$$\text{CrossAtten}(Q_C, K_C, V_C) = \text{Softmax} \left(\frac{Q_C K_C^T}{\sqrt{d_{k_c}}} \right) V_C \quad (10)$$

Among these, $Q_C = H_D W_q$, $K_C = H_W W_k$, $V_C = H_W W_v$, W_q , W_k , and W_v are learnable weight matrices. The output of

cross-attention is cross-modal feature $H_C \in R^{N \times d}$, which captures the interaction between driving behavior and weather conditions.

In addition, the study introduces learnable gating weights to achieve dynamic fusion, thereby balancing the contributions of driving features and cross-modal features.

$$g = \sigma(W_g [H_D; H_C] + b_g) \quad (11)$$

$$H_{fused} = g \odot H_D + (1 - g) \odot H_C \quad (12)$$

In this equation, σ represents the Sigmoid function, $\odot$ denotes element-wise multiplication, and W_g and b_g are learnable weight matrices and bias terms. The model employs a gating mechanism that enables it to adaptively learn the optimal weights of driving features and cross-modal features. This process facilitates more effective feature fusion.

Finally, in order to measure the riskiness of driving behavior, the hazard function, $H_{fused} \in R^{N \times d}$, is first mapped to N single values, z , through a fully connected layer. Then, z is compressed to the range of 0-1 through a sigmoid function to obtain the risk probability, P . The formula is expressed as follows:

$$p = \sigma(H_{fused}W + b) \quad (13)$$

In this case, w is the weight vector and b is the bias term. The classification results for driving behavior are as follows:

$$\hat{y} = \begin{cases} 1 & \text{if } p \geq \tau \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\hat{y}$ denotes the binary classification result and τ represents the threshold. When the probability exceeds the established threshold, the result of the driving behavior recognition is deemed to be risky. Subsequently, all risky driving behaviors that occur during the entire time series are documented in the result $\hat{P}$, which contains all risky driving behaviors identified during the entire trip.

E. Loss Function and Training

To optimize model performance, the following loss function and training strategy were adopted. The model loss $\mathcal{L}_{pred}$ is the predicted cross-entropy loss:

$$\mathcal{L}_{pred} = \frac{1}{N} \sum_{n=1}^N \text{BCE}(y_n, \hat{y}_n) \quad (15)$$

Among these, BCE is the binary cross-entropy loss, which is used to measure the difference between the model's predicted results and the actual labels.

During the training phase, the Adam optimizer is employed to update the model parameters, and an early stopping strategy is implemented to avert overfitting. Specifically, the model's performance is monitored on the validation set. In the event that the validation set loss does not decrease within a pre-defined number of epochs, the training process is terminated. Furthermore, a three-fold cross-validation procedure is implemented during the training phase to enhance the model's generalization capability, thereby facilitating its adaptation to the individual characteristics of diverse drivers.

F. Statistical Significance Testing

When employing J-fold cross-validation, standard paired t-tests tend to underestimate the variance due to overlapping training sets, leading to overly "optimistic" results. To achieve more robust significance inferences, this study adopts the modified paired t-test proposed by Nadeau & Bengio [42]. Specifically, for each baseline M_n and the proposed model M_0 , performance measure vectors $P_n = [p_{n0}, p_{n1}, \dots, p_{nn}]$ and $P_0 = [p_{00}, p_{01}, \dots, p_{0n}]$ are computed on the same validation fold for each fold $j = 1, \dots, J$. The absolute difference between them is calculated as $d_j = |P_0 - P_n|$, where the operator $(-)$ denotes element-wise subtraction. With $\bar{d} = \frac{1}{J} \sum_j d_j$ and $S_d^2 = \frac{1}{J-1} \sum_j (d_j - \bar{d})^2$, the Nadeau-Bengio variance inflation correction is applied to account for the positive correlation induced by overlapping training sets:

$$\widehat{Var}(\bar{d}) = \left(\frac{1}{J} + \frac{n_2}{n_1} \right) S_d^2 \quad (16)$$

where n_1 and n_2 represent the number of training and test samples per fold, respectively. From this modified variance, as proposed by Nadeau et al., the t-statistic is derived as:

$$t = \frac{\bar{d}}{\sqrt{\widehat{Var}(\bar{d})}} \quad (17)$$

where $\bar{d}$ denotes the arithmetic mean of the differences. Although this method has been previously utilized and evaluated by authors [42], [43] our model comparisons are not solely predicated on this modified test statistic. Results are reported with confidence intervals, which are also sufficient for comparing model performance.

V. EXPERIMENTAL RESULTS

A. Datasets

The dataset used in this experiment was obtained from the 7th "Teddy Cup" Data Mining Challenge. The dataset under consideration includes two types of data: vehicle trajectory data and weather data. The vehicle trajectory data set includes the driving trajectories of 450 commercial transportation vehicles from August to October 2018, with each vehicle having approximately 15 trip records. These trajectories were mainly collected from vehicles operating on highways and urban arterial roads, reflecting long-distance and continuous driving scenarios rather than dense urban intersections. Vehicle operating information was recorded by sensors, including operating status, speed, angle, longitude, latitude, time, and mileage. The volume of vehicle trip data ranges from 75,169 to 84,411 records. The weather data set includes temperature, wind direction, wind speed, and 12 types of weather for the corresponding time and region. The dataset is partitioned into training, validation, and test sets with a ratio of 7:1:2, respectively.

B. Experimental Setup

1) *Implementation Details*: CAFFT was implemented using PyTorch and Python. All experiments were conducted on an Intel Core i5-12600K, 32 GB RAM, and an Nvidia RTX 4090

TABLE I
THE OPTIMAL MODEL HYPERPARAMETER SETTING

	Hyperparameter	Value
Transformer	Emdedding dimension	128
	Number of ProbSparse Attention layers	2
	Number of Window Attention layers	1
	Number of Attention heads	4
	Top_u	5%
Model training	Batch size	32
	Maximum epochs	100
	Random seed_cv	42
	Early-Stopping patience	10

GPU. We trained our method using the Adam optimizer. The learning rate was initially set to $5e-4$, with a weight decay of $1e-5$. Training was conducted for 100 epochs, employing an early stopping strategy to prevent overfitting: model performance was monitored on the validation set, and training ceased if the validation loss did not decrease within 10 epochs. The hyperparameters for the optimal model are configured as follows TABLE I:

2) *Baseline*: We compare CAFFT against two categories of baseline methods: 1) traditional machine learning approaches, namely One-Class SVM (OCSVM) and Naive Bayes Classifier (NBC) and 2) deep learning methods, including LSTM, SLAFusion, Informer, and TFT:

OneClassSVM [15]. Compared to SVM, OneClassSVM can effectively handle scenarios with small sample sizes, such as separating dangerous behaviors in a single trip.

Naive Bayesian Classifier(NBC) [16]. This classifier is based on Bayesian theory and classifies samples according to the probability that they belong to each category.

Attention fusion based on SAX and LSTM method (SLAFusion) [44]. This method constructs a deep neural network that combines SAX and LSTM, and introduces an attention mechanism to learn and detect risky driving behavior from heterogeneous multi-source data.

Informer [31]. As an innovative time series forecasting model, Informer efficiently captures long-term dependencies in time series data through its design for long sequence inputs. It can be effectively applied to the identification of dangerous driving behaviors that require analysis of extended time series data.

TFT [45]. The Temporal Fusion Transformer (TFT) is a model specifically designed for multivariate time series forecasting. By integrating self-attention mechanisms and gated residual networks, it can handle complex patterns and provide interpretable predictions. This makes it suitable for identifying dangerous driving behaviors from multi-source heterogeneous sensor data.

C. Evaluation Mechanism

For evaluating these methods, we employ a precision-recall framework. Recall and precision are calculated according to

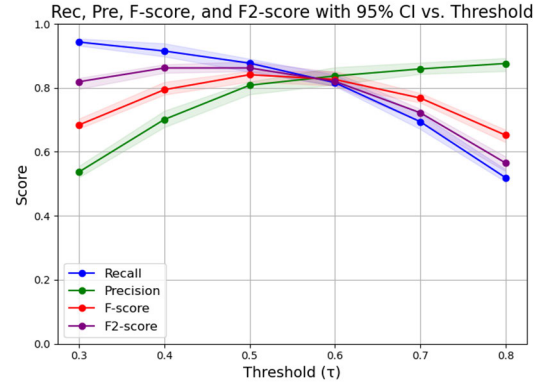


Fig. 7. Model Performance at Various Thresholds (The shaded regions indicate the 95% confidence intervals).

Equations (18) and (19):

$$Rec = \sum_{\hat{X}_i = \tilde{P}_i} \tilde{P}_i / \sum_i \tilde{P}_i, \quad (i = 1, 2, \dots, N_P) \quad (18)$$

$$Pre = \sum_{\hat{X}_i = \tilde{P}_i} \tilde{P}_i / \sum_i \hat{X}_i, \quad (i = 1, 2, \dots, N_P) \quad (19)$$

where N_P represents the total number of data records contained in a trip P , and $\hat{X}_i = \tilde{P}_i$ represents the i_{th} extracted driving behavior that was detected as risky driving behavior and is actually risky.

Given that the cost of false negatives in safety-critical scenarios is significantly higher than that of false positives, we further report F_β , where $\beta > 1$, to increase the weight of recall. This configuration places a greater emphasis on the model's performance in identifying true dangerous driving behaviors, thereby enhancing the model's effectiveness in practical applications. The formula for F_β is as follows:

$$F_\beta = \frac{(1 + \beta^2) \cdot Precision \cdot Recall}{(\beta^2 \cdot Precision) + Recall}, \quad \beta \in \{1, 1.5, 2\}. \quad (20)$$

When reporting experimental results, we selected $\beta = 2$, which aligns with the application objective of "minimizing false negatives in safety-critical scenarios." Concurrently, we also report the standard F1-score in our results, defined by the following formula:

$$F - score = \frac{2 * Pre * Rec}{Pre + Rec} \quad (21)$$

D. Models Performance Comparison

1) *Threshold Analysis*: In this study, the decision threshold τ is determined on the validation set and subsequently applied to the test set. To analyze model performance under various threshold values, we conducted experiments with $\tau \in (0.3 - 0.8)$. The experimental results, presented in Fig. 7, indicate that at $\tau = 0.5$, the model achieves its maximum F-score while maintaining an F2-score close to its maximum, thereby striking a balance between precision and recall. Accordingly, we set the threshold to $\tau = 0.5$.

TABLE II

RESULTS OF DIFFERENT RISKY DRIVING BEHAVIOR RECOGNITION METHODS. (RESULTS WITH 95% CI AND FINAL RESULTS ARE RETAINED TO 3 DECIMALS WITHOUT ROUNDING DURING THE ACTUAL EXPERIMENT)

<i>Method</i>	<i>Rec</i> (95%CI)	<i>Pre</i> (95%CI)	<i>F-score</i> (95%CI)	<i>F2-score</i> (95%CI)	<i>PR-AUC</i> (95%CI)
OneClassSVM	0.787(0.782–0.793)	0.774(0.748–0.801)	0.781(0.770–0.792)	0.785(0.757–0.812)	0.803(0.784–0.822)
NBC	0.655(0.618–0.692)	0.648(0.581–0.716)	0.652(0.614–0.689)	0.654(0.631–0.677)	0.756(0.730–0.782)
LSTM	0.833(0.827–0.840)	0.788(0.777–0.798)	0.81(0.803–0.816)	0.824(0.820–0.827)	0.809(0.788–0.830)
SLAFusion	0.847(0.838–0.855)	0.783(0.75–0.815)	0.813(0.806–0.826)	0.833(0.818–0.848)	0.839(0.825–0.854)
Informer	0.848(0.828–0.868)	0.803(0.783–0.823)	0.825(0.820–0.830)	0.839(0.827–0.851)	0.855(0.835–0.876)
TFT	0.857(0.846–0.869)	0.802(0.747–0.857)	0.829(0.815–0.842)	0.846(0.825–0.867)	0.873(0.857–0.889)
CAFFT	0.874(0.856–0.891)	0.799(0.768–0.831)	0.835(0.828–0.841)	0.858(0.840–0.875)	0.892(0.880–0.903)

TABLE III

F2-SCORE STATISTICAL SIGNIFICANCE TESTING RESULT

<i>Method</i>	<i>F2-score</i>	<i>t-score</i>	<i>p-value</i>
CAFFT	0.858	—	—
OneClassSVM	0.785	10.2427871	0.00E+00
NBC	0.654	51.88659509	0.00E+00
LSTM	0.824	3.407580834	3.35E-05
SLAFusion	0.833	13.75275092	0.00E+00
Informer	0.839	4.755563543	3.1177E-07
TFT	0.846	4.661244034	4.6116E-07

2) *Experimental Results*: For experimentation, the dataset was divided into three relatively independent experimental subsets, corresponding to vehicles numbered 001-150, 151-300, and 301-450, respectively. This division facilitates subsequent validation of the proposed method’s universality and stability. When presenting experimental results, we report only the averaged results across these three experimental groups, as shown in TABLE II. Based on these experimental findings, the following conclusions can be drawn.

Firstly, the proposed CAFFT method demonstrates the best performance. This indicates that the multi-granularity driving behavior extraction and subsequent result fusion mechanism is more effective in preventing misclassifications and missed detections.

Secondly, due to the imbalanced distribution of positive and negative samples within the dataset—for instance, only 1.48% of the summarized data generated by vehicle “AB00257” was marked as dangerous—the interpretability of the Precision (Pre) metric is relatively low. This is also one of the reasons why we set the parameter $\beta > 1$ in the F_β -score calculation formula.

Consequently, the Informer model’s advantage in Pre does not necessarily imply its superiority over CAFFT. Overall, the proposed CAFFT method exhibits superior performance compared to the aforementioned baseline methods.

3) *Statistical Significance Testing Results*: The statistical test results indicate that the improvement in F2-Score

demonstrated by CAFFT, relative to the comparison models, is statistically significant. The T-score and P-value obtained from the Nadeau–Bengio corrected paired t-test, as shown in TABLE III, confirm that the performance differences between the methods remain robust even after controlling for the variance underestimation caused by overlapping training sets. This suggests that the model proposed in this study possesses a consistent advantage in the objective of “recalling higher risk samples,” and its improvement is not due to random fluctuations but rather stems from systematic performance enhancements derived from the model architecture and fusion strategy. Given that F2-Score places greater emphasis on recall, these results validate the proposed method’s superior capability in suppressing false negatives within high-risk behavior identification scenarios.

4) *β -Sensitivity Analysis*: To confirm the reliability of our conclusions when increasing the recall weight by setting $\beta > 1$, we conducted a sensitivity analysis on $\beta \in \{0.5, 1, 1.5, 2, 3\}$, covering different false negative (FN) to false positive (FP) cost trade-offs. For each β , we report the F_β -score along with its 95% CI. The results, shown in Fig. 8, demonstrate that in the recall-priority range where $\beta \geq 1$, the advantage of our method over the baselines remains stable, indicating the robustness of our conclusions to the choice of β . This aligns with the application objective of “prioritizing the reduction of false negatives in safety-critical scenarios.” Accordingly, we designate $\beta = 2$ as our primary evaluation metric.

E. Ablation Experiments

1) *Sub-Module Ablation Experiments*: In order to verify the impact of each submodule in the CAFFT model on the identification results of risky driving behavior, an ablation experiment was conducted. Initially, the driving process was not divided during data preprocessing to verify the impact of driving process extraction on the experimental results, which was represented as CAFFT (- D_p). Secondly, the hierarchical encoder structure employed in the method was substituted with a non-hierarchical structure, which was denoted as CAFFT (-Encoder). This modification was implemented to ascertain the efficacy of the hierarchical encoder structure in capturing global information. Thirdly, the weather module, which learns

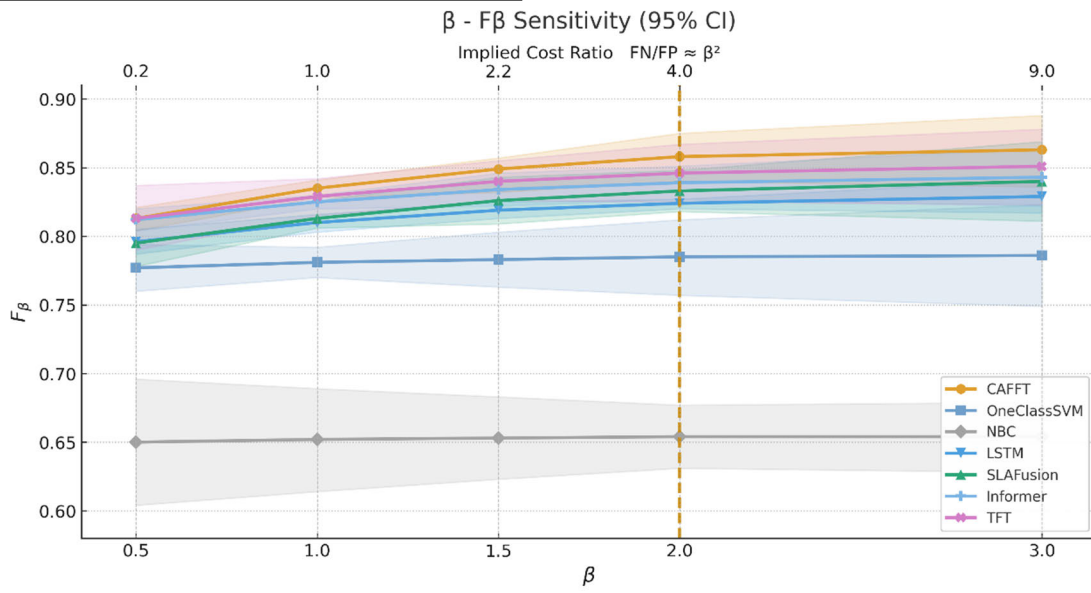


Fig. 8. β -Sensitivity Analysis with 95% CI. This plot illustrates the performance (F_β -score) of different models. The shaded regions indicate the 95% confidence intervals. The implied cost ratio ($FN/FP \approx \beta^2$) is also shown.

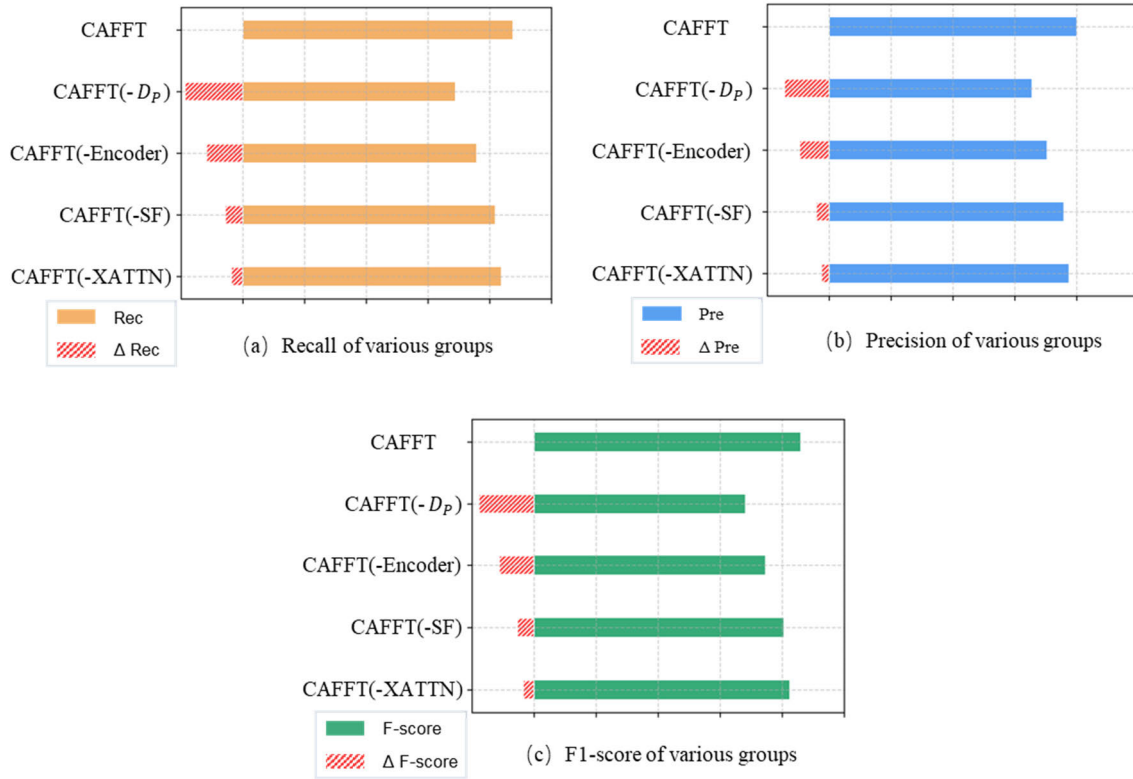


Fig. 9. Results of each group in the ablation experiment. Δ indicates the differences between each experimental group with CAFFT.

situational factors, was removed and represented as CAFFT (-SF) to demonstrate that situational factors have a significant impact on the recognition results. Finally, the attention mechanism in the cross-modal dynamic fusion module was removed to verify the impact of the cross-attention mechanism on the experimental results, and it was represented as CAFFT (-XATTN). Specifically, rather than employing the

cross-attention mechanism to associate driving behavior features and weather features, the driving behavior features and weather features were directly concatenated as input for subsequent steps.

A subsequent analysis of the results of the ablation experiment, as illustrated in Fig. 9, yielded the following conclusions. First, it is necessary to extract the driving

TABLE IV
ABLATION EXPERIMENTS RESULTS FOR WEATHER SUB-MODALITIES

setting	Rec	Pre	F-score	F ₂ -score	PR-AUC	ΔF_2
CAFFT	0.874	0.799	0.835	0.858	0.684	—
CAFFT(-WT)	0.823	0.768	0.795	0.811	0.679	-0.047
CAFFT(-TMP)	0.869	0.789	0.827	0.852	0.682	-0.006
CAFFT(-WS)	0.866	0.791	0.827	0.850	0.683	-0.008
CAFFT(-WD)	0.871	0.787	0.827	0.853	0.68	-0.005
CAFFT(+WT)	0.865	0.789	0.825	0.849	0.681	-0.009
CAFFT(+TMP)	0.819	0.764	0.791	0.807	0.677	-0.051
CAFFT(+WS)	0.821	0.767	0.793	0.810	0.679	-0.048
CAFFT(+WD)	0.817	0.759	0.787	0.805	0.675	-0.053

process before identifying driving behavior. The classification of driving behavior based on fine-grained criteria has been demonstrated to be more effective in identifying risky driving behavior. Consequently, CAFFT exhibits superior performance in comparison to CAFFT (- D_P). Secondly, the results of the original model were superior to those of CAFFT (-SF), indicating the necessity of considering situational factors when identifying risky driving behavior. Furthermore, situational factors exerted the greatest influence on identification results in all verifications. Finally, both CAFFT (-XATTN) and CAFFT (-Encoder) demonstrate a lower level of performance in comparison to the original model, thereby providing a comprehensive validation of the impact of the cross-attention mechanism and the hierarchical encoder structure on experimental outcomes.

2) *Ablation Experiments of Different Weather Sub-Modalities*: Furthermore, to investigate the independent contributions and complementary effects of individual sub-modal features within weather information on dangerous driving behavior identification performance, this study conducted additional ablation experiments. Specifically, two complementary ablation strategies were designed. The first strategy involved removing a single sub-modality: one by one, weather type (WT), temperature (TMP), wind speed (WS), and wind direction (WD) sub-modalities were removed while keeping other weather features constant. The second strategy involved adding only a single sub-modality: only one weather feature was retained, with all others removed. All other experimental settings remained consistent with the original model to ensure comparability of results. This yielded a total of eight model variants:

CAFFT(-WT), CAFFT(-TMP), CAFFT(-WS), CAFFT(-WD), CAFFT(+WT), CAFFT(+TMP), CAFFT(+WS), and CAFFT(+WD).

The experimental results are presented in TABLE IV. When employing the “remove single sub-modality” strategy, model performance degraded in all scenarios, indicating that each sub-modality provides complementary contextual information for dangerous driving identification. Among these, the removal of weather type information led to the most significant performance drop ($\Delta F_2 = 0.047$), followed by wind speed

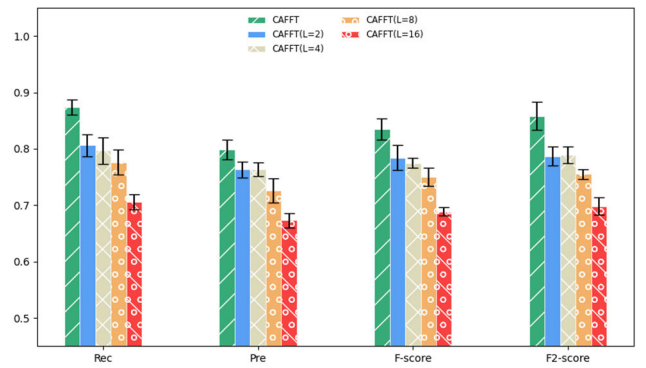


Fig. 10. Experimental results under different grain size window divisions. The error bars in the figure represent 95% (CI) error bars.

($\Delta F_2 \approx 0.008$) and temperature ($\Delta F_2 \approx 0.006$), with wind direction having the least impact ($\Delta F_1 = 0.005$).

Under the “add only single sub-modality” strategy, the model retaining only weather type, CAFFT(+WT), performed best ($F_2 = 0.849$), suggesting that categorical weather conditions (e.g., clear, rainy, snowy, foggy) provide relatively direct indications of driving risk. Conversely, models containing only temperature or wind-related features (CAFFT(+TMP), CAFFT(+WS), CAFFT(+WD)) exhibited limited identification capability ($F_2 \approx 0.805$ – 0.810), indicating that continuous meteorological variables alone are insufficient to comprehensively characterize the impact on the driving environment.

F. Granular Analysis of the Driving Process

This section undertakes an analysis of the impact of different time granularity time window extraction processes on recognition results. CAFFT (L) denotes the results obtained when only one process extraction is performed on driving behavior for different window sizes, where L is the window size.

As illustrated in Fig. 10, while the results obtained by CAFFT (L=4) exhibit slight differences compared to those obtained by CAFFT (L=2), the difference is not significant. However, under the L=4 granularity division, the dimension of the feature vector is only half that of the L=2 granularity

TABLE V
MODEL COMPLEXITY AS A FUNCTION OF WINDOW SIZE

Window L	Step S	Parameters(M)	GPU Memory(GB)	Inference Latency / Batch (ms)	F2-score
2	2	6.28	4.1	35.7	0.787
4	4	6.28	3.2	37.6	0.789
8	8	6.28	4.5	48.3	0.755
16	16	6.28	4.9	65.4	0.698

TABLE VI
CROSS-TEST SET GENERALIZATION TEST RESULTS

<i>F-score</i>	CAFFT(a)	CAFFT(b)	CAFFT(c)
Test set 1	0.851	0.847	0.854
Test set 2	0.855	0.857	0.861
Test set 3	0.862	0.865	0.865
Averages	0.856	0.856	0.860
Standard deviation	0.004849487	0.007350344	0.004591354

division, which reduces the computational load of the model as TABLE V show. This is the primary rationale behind the selection of $L=4$ as the primary division in the final model, as opposed to $L=2$. Furthermore, the experimental results obtained under the $L=16$ granularity division differ significantly from those obtained under $L=8$. This finding indicates that dividing the driving process into overly long-time windows does not align with the actual duration of a single driving behavior in the real world. Consequently, this misalignment affects the final experimental results. Finally, CAFFT exhibited a marked superiority over those of the other control groups, thereby substantiating the necessity of multi-granularity segmentation of the original trajectory data as opposed to single-granularity segmentation.

G. Cross-Test Set Generalization Test Experiments

In order to ascertain the efficacy of the proposed methodology in adapting to the personalized differences among drivers, cross-test set generalization test experiment were conducted on models trained using different training sets. The experimental results are presented in TABLE VI. In this table, CAFFT (a) represents the model that was trained on test set 1 (numbers 001-150), CAFFT (b) represents the model that was trained on test set 2 (numbers 151-300), and CAFFT (c) represents the model that was trained on test set 3 (numbers 301-450). The experimental results demonstrate that our method generates consistent test results on experimental sets comprising various drivers, with the standard deviation of the F-score for the three models in different experimental sets being less than 0.007. This finding indicates that the CAFFT model exhibits particular strengths in adapting to the individual differences among different drivers, effectively accommodating these variations, and meeting the requirements of real-world driving behavior analysis and application.

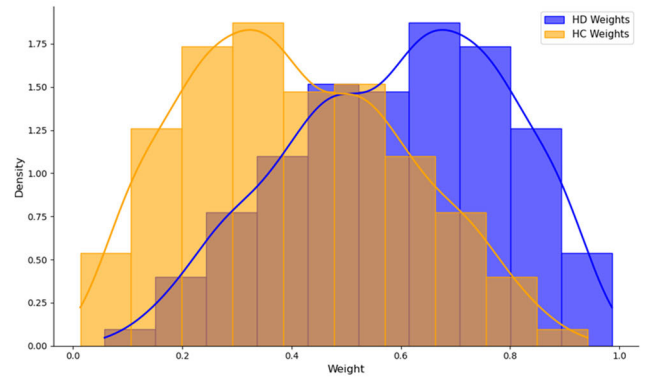


Fig. 11. Gated Weight Distribution. The figure displays histograms and kernel density estimates of the H_D Weights (blue), and H_C Weights (orange), after model training.

H. Gated Weights Visualization

In this subsection, we visualize the distribution of gated weights for driving feature H_D and cross-modal feature H_C . This visualization not only helps to verify whether the proposed gating mechanism aligns with the intended modeling logic but also provides transparency and interpretability for understanding model decisions. Furthermore, it elucidates how the model leverages core information from different modalities to accomplish the final classification in dangerous driving behavior recognition.

In Fig. 11, the gated weights for the driving feature H_D exhibit higher concentration and a right-skewed distribution, with peak weights predominantly falling within the 0.55–0.80 range. This indicates that the model tends to enhance trajectory dynamic information across most feature dimensions. Conversely, the gated weights for the cross-modal feature H_C , after integrating the weather modality, are generally skewed towards the 0.10–0.45 range, displaying a flatter density variation.

This suggests that the contribution of weather to dangerous driving identification has stronger dimensional heterogeneity, with high weights assigned only in a few dimensions. The overlap between the two distributions is primarily concentrated in the 0.35–0.55 range, representing a synergistic strengthening effect between trajectory and weather in these channels.

From an overall distribution perspective, the distribution of H_D is sharper, while that of H_C is more dispersed. This suggests that the trajectory modality demonstrates more stable consistency during the cross-modal fusion process, whereas the contribution of the weather modality exhibits greater inter-dimensional fluctuation. This result validates CAFFT's design hypothesis: the primary discriminative information for dangerous driving behavior identification originates from continuous trajectory dynamics, while weather, as a contextual factor, primarily exerts a "modulatory" influence, significantly affecting the fusion outcome only in specific dimensions.

VI. CONCLUSION AND DISCUSSION

In light of the numerous challenges associated with identifying hazardous driving behaviors, this paper puts forth a Transformer-based cross-attention feature fusion dangerous driving behavior recognition model, designated as CAFFT. Additionally, it assesses the efficacy of the model on a real dataset.

The CAFFT model is composed of three primary modules: a driving sequence feature learning module, a weather feature learning module, and a cross-modal dynamic fusion module. In the driving sequence feature learning module, firstly, to circumvent the issue of misjudged results in the driving process segmentation, disparate sizes of sliding time windows are employed for segmentation. Secondly, to capture the long-term dependencies and detailed information of the driving behavior, a hierarchical Transformer encoder structure is designed, where the ProbSparse Attention mechanism is introduced in the global encoder to capture the long-term dependencies of the driving behavior, and the window Attention mechanism is introduced in the local encoder to capture the fine-grained fluctuations. In the weather feature learning module, an effective extraction method was designed to identify weather features related to driving behavior. This method combines One-Hot encoding, numerical normalization, and a one-dimensional convolutional network. In the cross-modal dynamic fusion module, we also fuse driving behavioral features with weather features through a cross-attention mechanism to characterize the effect of situational factors on driving behavior during the vehicle's driving process. Concurrently, in order to enhance the model's capacity to adapt to the distinct characteristics of different drivers, we implemented three-fold cross-validation during the training process to augment the model's generalization capability. We subsequently verified the model's ability to adapt to individualized differences among different drivers through cross-test-set generalization tests.

To this end, experimental validation on a real driving dataset is conducted to evaluate the performance of the proposed method through a series of experiments, including model

performance validation, ablation experiments, driving process extraction granularity analysis experiments, and cross-test-set generalization test experiments. In comparison with benchmark methods, CAFFT achieved an F2-score of 0.858 (95% CI: 0.840-0.875) and a PR-AUC of 0.892 on a real-world driving dataset. This represents a 1.2-1.9 percentage point improvement in F2-score over powerful Transformer-based models such as Informer and TFT. Statistical significance analysis using the Nadeau-Bengio corrected paired t-test confirmed that these improvements are not attributable to random variability (p-values $<1e-4$ for all major baselines). These results indicate that the proposed multi-granularity hierarchical encoding and cross-modal fusion of weather information effectively reduce false negatives in risky driving behavior identification. The performance of each component of the model was verified by means of ablation experiments. In the granularity analysis experiment, the performance of the model at different granularities is verified. In the cross-test set generalization test experiments, the CAFFT model demonstrated a superior capacity to adapt to the driver's unique characteristics.

Future research can be conducted in the following areas. Firstly, risky driving behavior recognition is inherently scenario-dependent, as identical driving maneuvers may correspond to different risk levels under varying operational conditions and human response capabilities. Therefore, further investigation of contextual factors and their combined effects is necessary to better reflect real-world driving scenarios. Secondly, the personal physiological and psychological factors of drivers also affect driving behavior performance. A promising avenue for enhancing the recognition performance of driving behavior involves the integration of these heterogeneous data with existing research. This necessitates interdisciplinary research, such as the integration of psychology and physiological information disciplines. Thirdly, given that dangerous driving behavior recognition is a research topic that contributes to driving safety, active intervention based on recognition results is a necessary research direction. Subsequent research endeavors may explore the development of methods to proactively intervene in drivers, with these interventions being informed by the findings from driving behavior recognition. Finally, Given the critical nature of road traffic safety, with consequences that are often life-or-death, the timely application of the researched technology is of the utmost importance. However, this assertion is subject to various influences and challenges, including the maturity and reliability of the technology, the system's resilience to attacks, and the efficacy of data protection mechanisms. Addressing these issues can more effectively prevent and handle traffic accidents, further improving road traffic safety.

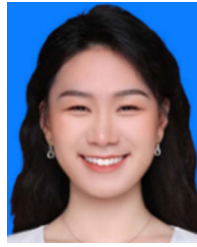
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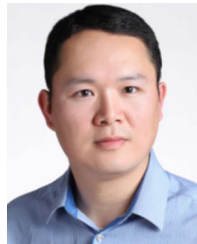
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